# MIPS Control Panel Timing Generator

Overview
December 3, 2018
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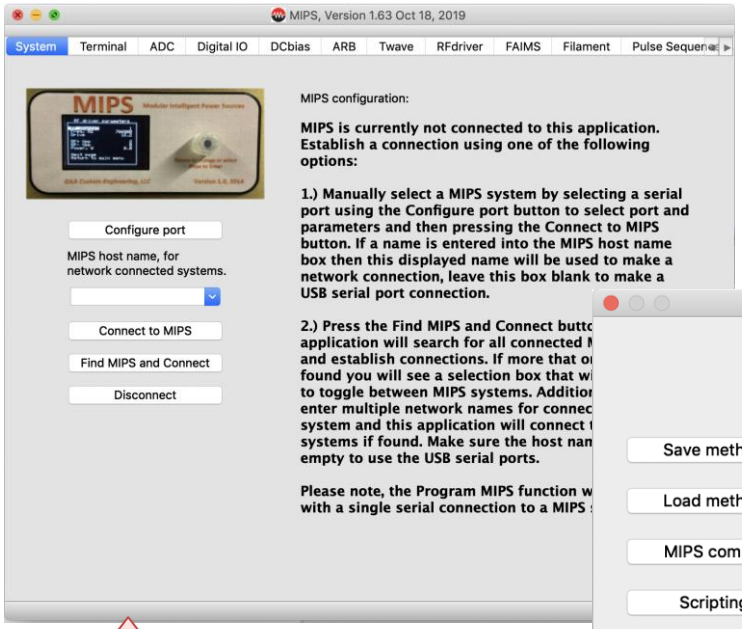
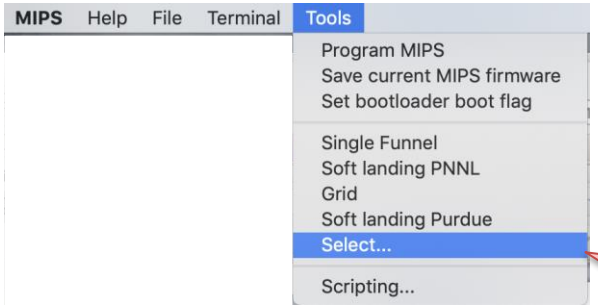
Revised March 21, 2026

# Loading MIPS Control Panels

## MIPS main startup dialog

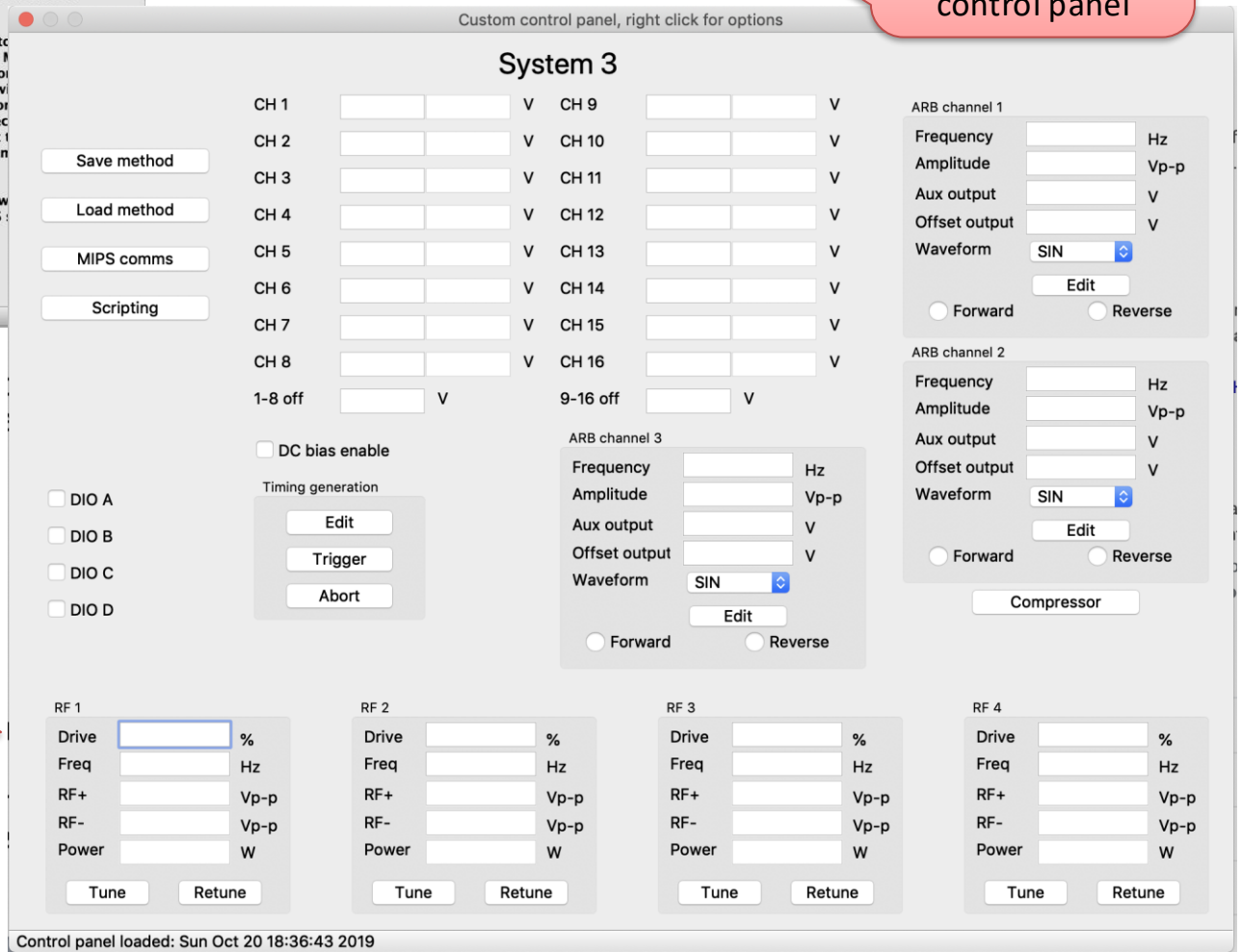



MIPS menu

Select this option to select a .cfg file that will load a control panel

MIPS host application, supports PC, Mac and Linux

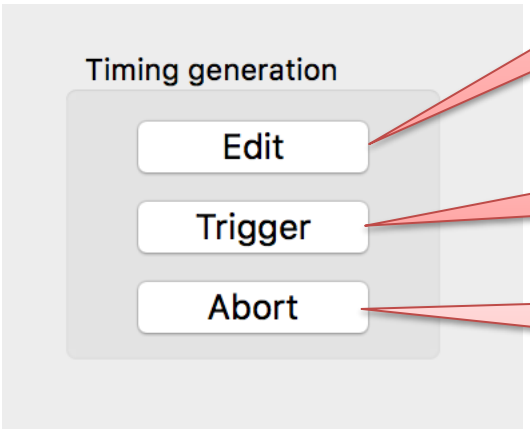



Example control panel, these are easy to customize based on your system. When the control panel loads the MIPS main dialog will minimize.

# Control panel Interface

Adding the following command to the control panel configuration file will enable the timing generation function discussed in this document:
TIMING,Timing generation,MIPSname,X, Y
MIPSname is the name of the MIPS system where this timing control will be applied. X and Y define the location of the control.

This is the dialog you will see on your control panel.




Press this button to popup a dialog box to allow editing the pulse sequence

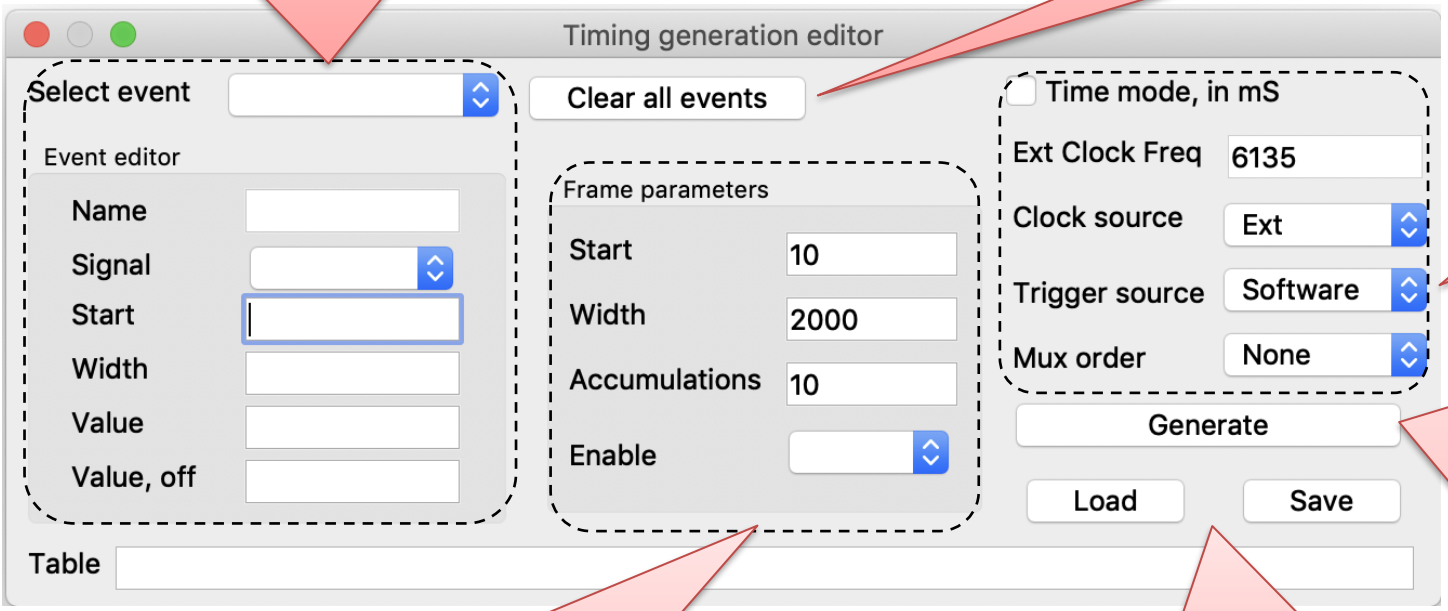
Press this button to trigger the pulse sequence and start a data acquisition

Press this button to stop an acquisition that is in progress.

# Pulse sequence editor

This section allows you to define an event in the pulse sequence. You can have as many events as needed in a pulse sequence.

This button will clear all events and allow you to start a new sequence.




This section defines clock and triggering options.

This button will use the current settings to generate a pulse sequence and display it in the table box.

This section defines the data collection frame, the number of accumulations and allows defining an Enable output signal if needed.

These buttons allow you to load and save pulse sequences to a data file.

# Pulse sequence editor

## Event definition

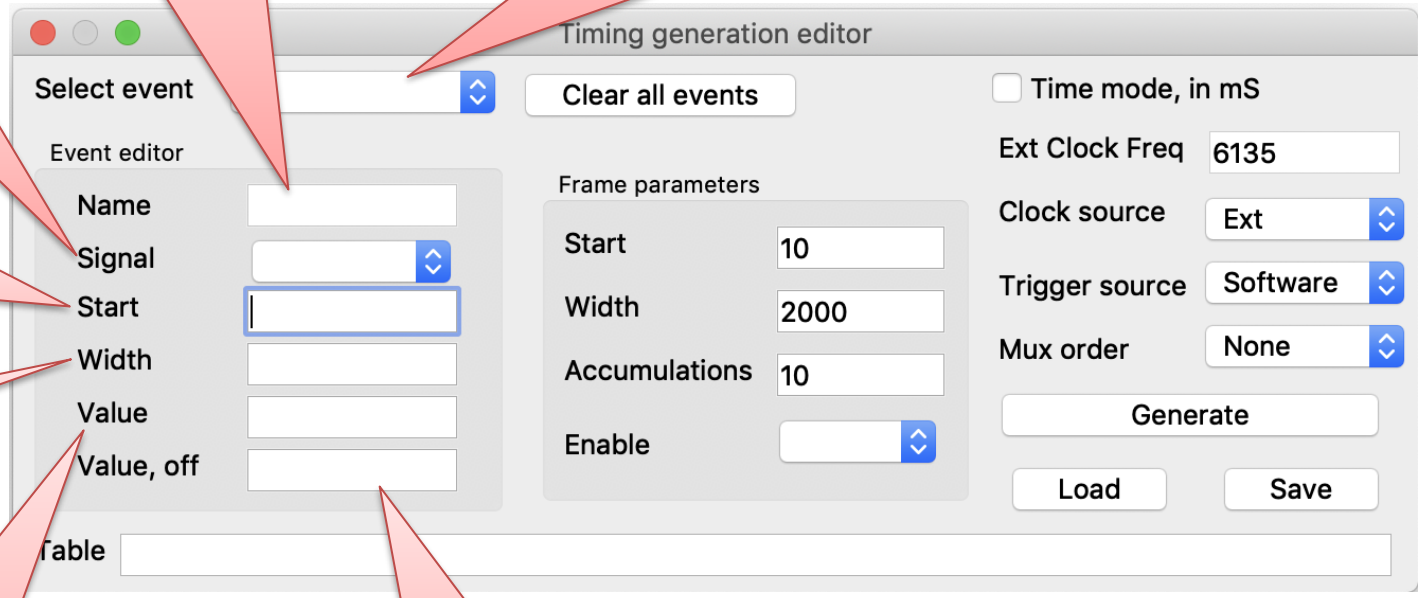
Name of the selected event.

Allows you to define a new event, select an existing event, or delete an existing event.

This is the signal this event defines. This list is automatically populated from the available options.

Starting point (in clock cycles) for this event in the pulse sequence.

Width (in clock cycles) of this event. This width can be set to 0 if you only wish to generate a start time point.




Output value for this event when it's active

Output value for this event when it's inactive

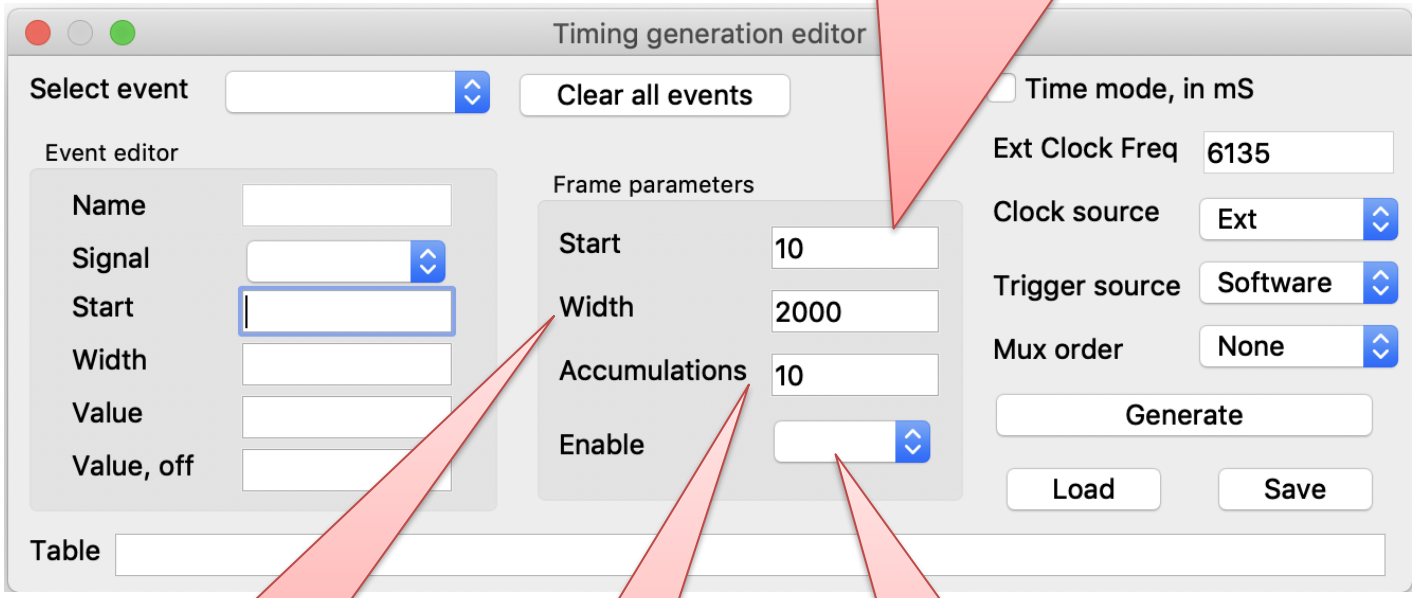
Define as many events as necessary. Multiple events can share the same start and width values, but their names must be unique.

# Pulse sequence editor

## Frame parameters

The Frame parameters define the total length to the frame and the number of cycles the pulse sequence will repeat.

Defines the start point (in clock cycles) of the data acquisition frame in the pulse sequence.




Data acquisition frame width (in clock cycles).

This is the number of accumulations or cycles for this pulse sequence.

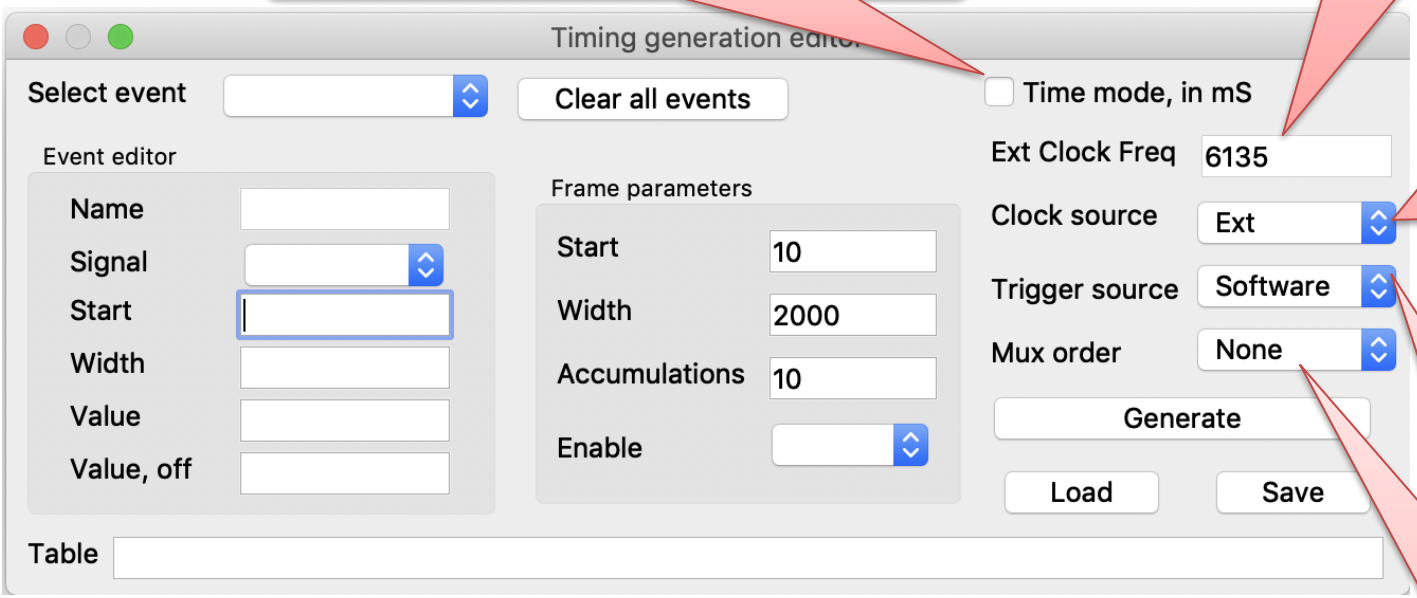
This option allows you to define an output logic signal that will indicate the acquisition interval.

# Pulse sequence editor

## Clock and trigger options

Pulse sequences can be defined in clock cycles or time in mS. Check this box if you wish to enter the parameters in mS.

If you have selected time mode and you're using an external clock, this box allows you to define the external clock frequency in Hz. This is needed to calculate the time in mS.




Defines the clock used by the pulse sequence generator:
Ext uses the Clk input.
ExtN uses the negative edge of the Clk input.
ExtS uses the S input.
There are also a number of internal clock frequency options.

Define the trigger option used to start a pulse sequence. Options include:
Software
External Trg input on the Pos edge, Neg edge, or Edge for any edge.

The pulse sequence generator supports generation of Hadamard multiplexing bit sequences. Use this option to select the desired option.

Note: The count values in the start and width boxes of the events and frame parameters can either represent fixed numbers, such as total counts or time, or references to other event start and width values. For instance, if you define an event named ACC with a start count of 100, you can then define another event or frame parameter's start or width as 25 + ACC.Start. You can use both the plus (+) and minus (-) operators. This feature allows you to logically link events, so changing the value of one event will redefine other events in a logical manner for your application.

# Pulse sequence options

- The event Width parameter can be set to INIT or RAMP when using this interface to generate a sequence that ramps a voltage channel.
- The time point can be set to a negative value to flag it for time point value adjustment.
- The signal selection combo box includes Delta t; this option allows you to define the time point adjustment value.
- If no signal is selected for an event, you can use the event as a time reference for other events.
- Multiple events can have identical start and stop times.
- The full pulse sequence is saved with the control panel save method button. You can also use the Save button on the pulse sequence editor dialog to save only the pulse sequence.
- You cannot generate nested loops with this interface, but you can use it to generate and test the inner loop of a more complex design, thereby speeding up development.

# Example pulse sequence

- In this example we generate a generic pulse sequence. This sequence consists of four main events:
  - Fill time
  - Trap time
  - Release time
  - Inject time
- For each of these events we define a start and width time, as well as a signal to control. The signal is optional and it can be left blank. If the signal is selected, then its active and off values need to be defined.
- In this example we will focus on the event's start and width values. This sequence is defined in a way that the user only needs to edit the Inject time to define where in the sequence the injection occurs; all the events are linked. Below is a table of event start and width values to illustrate the capability:

| Event name | Start | Width |
|---|---|---|
| Fill time | Trap time.Start-Fill time.Width | 100 |
| Trap time | Inject time.Start-Trap time.Width | 10 |
| Release time | Inject time.Start | 20 |
| Inject time | 200 | 9 |

- In this example the user only needs to adjust the width values for the first three events and the Inject time Start. The signals being controlled are not shown here, as they will depend on your system. Note that Trap time does not control a signal; it represents a delay while the ions are held in the trapping region.

# Hadamard multiplexing

- Generation of a Hadamard multiplexing pulse sequence requires first the generation of the injection event. The following events must be defined as outlined in the example pulse sequence:
  - Fill time
  - Trap time
  - Release time
  - Inject time
- When the pulse sequence is generated all the injection times are calculated based on the bit order, then the above four events are used to calculate each injection event.
- Events defined in the pulse sequence with any other event names are used as defined and can control other aspects of the experiment.
- Please note, the Injection time event's start time must be set to zero.